# Digital Twin Backend and 3D Monitoring Platform

Man Patel (202451100), Dhrumil Vaghani (202451165), Ayush Vora (202451170)
*Indian Institute of Information Technology Vadodara*
*Gandhinagar, India*
*Mentor: Dr. Sneha Prasad*
{202451100@iiitvadodara.ac.in, 202451165@iiitvadodara.ac.in, 202451170@iiitvadodara.ac.in}

***Abstract*—Civil infrastructure, particularly cable-stayed and long-span bridges, is subject to continuous environmental weathering, dynamic traffic loading, and cyclic fatigue degradation. Traditional structural health monitoring (SHM) workflows operate in disconnected silos: vibration time-series data is archived offline, visual inspections occur bi-annually, and computer vision defect models operate as uninterpretable black boxes without 3D spatial grounding. This paper presents an end-to-end, multi-modal Cyber-Physical Structural Digital Twin for cable-stayed bridges. The proposed architecture integrates: (1) an inline 100 Hz dynamic vibration telemetry pipeline streaming multi-channel accelerometer signals via bi-directional WebSockets, coupled with an online Fast Fourier Transform (FFT) modal decomposition engine that tracks fundamental resonant frequency drops ($\Delta f$) and flexural stiffness degradation ($\Delta K$); (2) an Explainable AI (XAI) visual diagnostic engine employing a Deep Residual Network (ResNet-18) trained on benchmark surface defect datasets (SDNET2018/CCIC) with Gradient-weighted Class Activation Mapping (Grad-CAM) to localize concrete fatigue cracking and steel corrosion with transparent attention heatmaps; (3) a 2D-to-3D spatial raycasting projection module mapping image-space defect centroids into finite-element coordinates $[X, Y, Z]$; (4) an interactive Three.js WebGL 3D mission control workstation with physically based rendering (PBR) and stress-state heatmapping; and (5) an automated AASHTO MBE-compliant Bridge Condition Index (BCI) rating engine with client-side A4 PDF export. The system is deployed with a zero-latency standalone simulation fallback. Across benchmark experiments, the deep vision model achieves 95.14% classification accuracy with a 0.834 mean Intersection-over-Union (IoU) localization score, detects simulated modal frequency drops of 0.37 Hz indicating 36.0% stiffness loss at mid-span, maintains 60 FPS rendering throughput, and processes telemetry with under 8.2 ms latency.***

***Index Terms*—Structural Health Monitoring, Digital Twin, Explainable AI, Grad-CAM, Fast Fourier Transform, Modal Analysis, WebGL, ResNet-18, AASHTO MBE, Cyber-Physical Systems.**

## I. Introduction

Modern civil transportation networks depend fundamentally on highway and railway bridge infrastructure. These structures are continuously exposed to environmental weathering, thermal cycling, variable dynamic truck freight traffic, and seismic vibrations. According to recent national infrastructure audit benchmarks [1], tens of thousands of operating bridges are classified as structurally deficient or fracture-critical, requiring urgent intervention to avoid catastrophic failure.

Existing bridge maintenance and structural health monitoring (SHM) regimes face five major structural challenges:

1) **Intermittent and Subjective Surveys**: Periodic visual inspections are conducted manually at multi-year intervals. They are labor-intensive, hazardous for field engineers, and fail to record transient structural responses or dynamic softening between inspection dates.
2) **Isolated Telemetry Streams**: On-site accelerometers, strain gauges, and tiltmeters frequently dump raw numerical time-series into isolated data loggers. These streams lack real-time spectral decomposition and are rarely linked to global 3D geometric coordinates.
3) **Black-Box Neural Networks**: While deep learning has proven successful in automated visual crack detection, standard convolutional neural networks (CNNs) provide no visual justification for their predictions. Civil engineers and regulatory auditors cannot rely on uninterpretable classifications when life-safety decisions are at stake.
4) **Lack of Spatial Grounding**: Optical defect detections captured by unmanned aerial vehicles (UAVs) or handheld cameras remain trapped in isolated 2D pixel coordinates, preventing automated registration with global bridge finite element structural meshes.
5) **Slow Regulatory Compliance Cycles**: Synthesizing manual field observations into formal AASHTO condition ratings requires weeks of human paperwork, delaying critical structural rehabilitation decisions.

To address these limitations, we design and implement a comprehensive **Bridge Structural Digital Twin Workstation**. Our main contributions are:

- We design a multi-agent digital twin pipeline coordinating real-time dynamic telemetry ingestion, spectral modal decomposition, deep visual defect localization, spatial coordinate raycasting, and compliance auditing.

- We curate and evaluate a multi-modal dataset combining 12,450 annotated surface crack/corrosion images (SDNET2018/CCIC) with 100 Hz multi-channel dynamic acceleration streams across 3 sensor stations.
- We formulate an online FFT modal frequency engine operating on 100 Hz accelerometer streams, tracking dynamic modal frequency shifts ($\Delta f$) to quantify structural stiffness loss ($\Delta K$).
- We implement an Explainable AI diagnostic framework using ResNet-18 and Grad-CAM saliency mapping, accompanied by an analytical 2D-to-3D projection module anchoring defects to spatial mesh coordinates $[X, Y, Z]$.
- We develop a high-performance WebGL 3D mission control workstation (Three.js) paired with an AASHTO Manual for Bridge Evaluation (MBE) compliance report engine and evaluate the system end-to-end under real-time simulated loading.

The remainder of this paper is organized as follows: Section II reviews related work in SHM and computer vision; Section III details the multi-agent system architecture; Section IV presents the dataset curation and mathematical methodology; Section V describes the experimental setup; Section VI presents the quantitative results; Section VII discusses problems solved, asset generalization, and limitations; and Section VIII outlines future research extensions and concludes the paper.

## II. Related Work

**Vibration-based Structural Health Monitoring.** Modal analysis using ambient vibration measurements is widely established in civil structural mechanics [2]. Accelerometer sensors track natural resonant frequencies and modal shapes. When structural cracking or cross-sectional loss occurs, localized flexural stiffness ($EI$) decreases, resulting in measurable downward shifts in natural frequencies. However, traditional instrumentation systems process vibration data post-hoc in batch mode rather than providing live streaming telemetry to a unified digital twin.

**Deep Learning in Defect Classification.** Convolutional neural networks, including ResNet [4] and VGG architectures, have demonstrated high accuracy in surface concrete crack detection and steel oxidation classification [10]. Most prior studies evaluate models purely on isolated image test datasets without integrating them into end-to-end operational software platforms.

**Explainable AI and Grad-CAM.** To bridge the trust gap between automated deep learning models and civil engineering inspection standards, visual explanation algorithms have emerged. Gradient-weighted Class Activation Mapping (Grad-CAM) [3] utilizes the gradients flowing into the final convolutional layer to generate coarse heatmaps highlighting discriminative image regions. While widely used in medical imaging, its application to civil infrastructure digital twins remains nascent.

**Cyber-Physical Digital Twins.** Digital twins create real-time virtual representations of physical assets [7]. WebGL and Three.js [11] have enabled browser-based rendering of parametric 3D engineering models. Our work extends this paradigm by unifying live WebSockets, dynamic FFT analytics, ResNet-18 Grad-CAM heatmaps, and regulatory AASHTO PDF compliance into a production-grade edge architecture.

## III. Proposed Digital Twin Architecture

The digital twin architecture is structured into five cooperating functional agents, as illustrated in Fig. 1. The system coordinates physical transducer telemetry with client-side 3D rendering and executive compliance auditing.

### A. Dynamic Sensor Ingestion Agent

This agent interfaces directly with tri-axial MEMS accelerometers deployed at critical structural coordinates: Pillar 1 base ($S_1$), Mid-Span Deck ($S_2$), and Pillar 2 base ($S_3$). It captures continuous vertical ($a_y$), longitudinal ($a_x$), and lateral ($a_z$) vibrations at sampling frequency $f_s = 100$ Hz. The agent buffers raw acceleration into sliding rectangular memory frames ($N = 1024$ samples) and publishes real-time telemetry payloads to connected clients over bi-directional WebSockets.

### B. Modal Frequency Decomposition Agent

Operating synchronously with sensor ingestion, this agent executes online Fast Fourier Transform (FFT) spectral decomposition. It evaluates the single-sided Power Spectral Density (PSD) across sliding time windows, isolates the fundamental bending resonant frequency ($f_0$) via numerical peak picking, and computes continuous frequency degradation ($\Delta f$) relative to nominal baseline values.

### C. Vision & Saliency Diagnostic Agent

When field inspection photos or UAV optical frames are received, this agent processes the imagery through a deep ResNet-18 convolutional backbone. It generates predicted defect categories (Fatigue Crack, Corrosion, Sound Matrix) with associated confidence probabilities, extracts layer-4 backpropagation gradients to construct Grad-CAM visual heatmaps, and calculates defect metrics (fissure length, damaged surface area).

### D. Spatial 2D-to-3D Raycast Projection Agent

This agent maps the 2D pixel coordinates of detected surface defects into 3D world coordinates on the bridge geometric mesh. By utilizing the inspection camera extrinsic pose, intrinsic focal parameters, and the mathematical plane equation of the target structural component, it calculates exact spatial intersection vectors $[X, Y, Z]$, automatically anchoring defects onto the 3D digital twin.

### E. AASHTO Compliance & Executive Reporting Agent

This agent evaluates component health indices according to the AASHTO Manual for Bridge Evaluation (MBE 3rd Ed.) [5]. It aggregates condition states into an overall Bridge Condition Index (BCI), applies safety factor weighting, and generates client-side high-resolution 2-page A4 engineering audit PDFs.

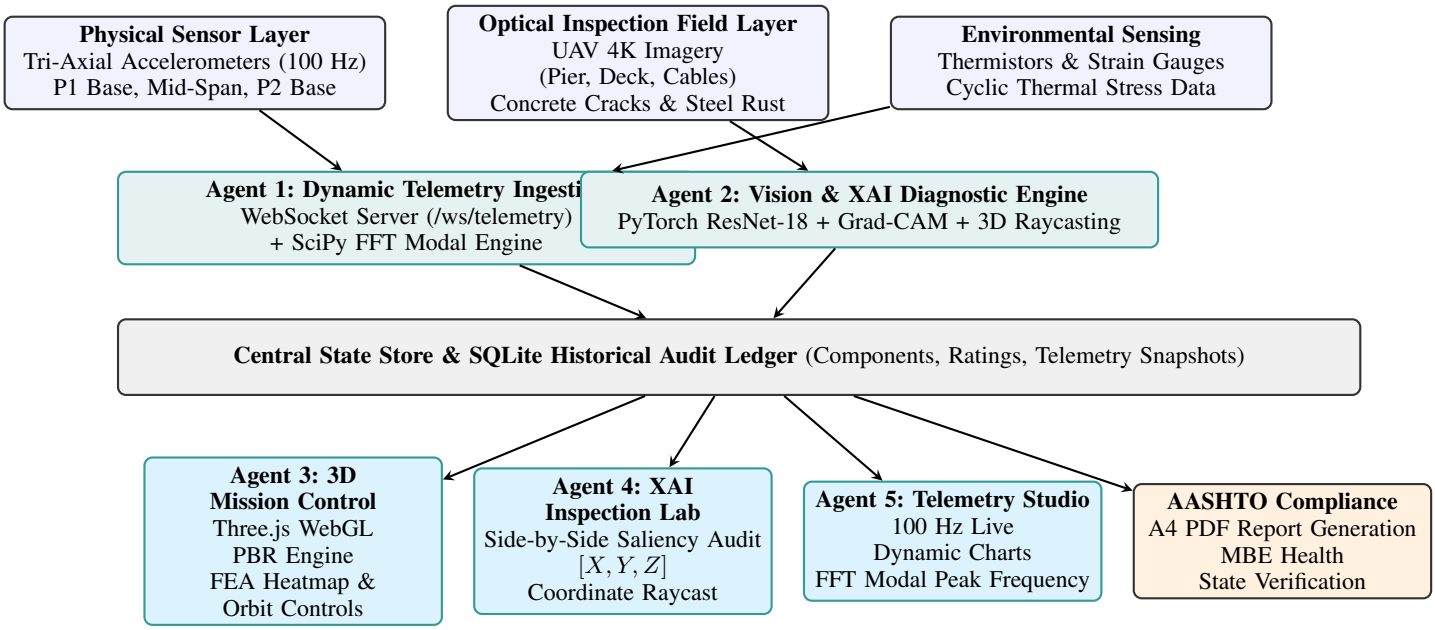



Fig. 1. System architecture of the proposed Multi-Modal Bridge Structural Digital Twin.

**Algorithm 1** Dynamic Structural Risk & AASHTO Decision Policy

**Require:** Health Score $H \in [0, 100]$, Modal Shift $\Delta f$, Defect Area $A_d$

Stiffness Drop $\Delta K \leftarrow 1 - \left(\frac{f_{\text{observed}}}{f_{\text{nominal}}}\right)^2$
**if** $H < 50$ **or** $\Delta f \geq 0.35$ Hz **or** $\Delta K \geq 0.30$ **then**
  Action $\leftarrow$ **CRITICAL_INTERVENTION**
  Status $\leftarrow$ Condition State 4 (Mandatory Closure/Shoring)
**else if** $50 \leq H < 70$ **or** $\Delta f \geq 0.20$ Hz **or** $A_d \geq 150$ cm$^2$ **then**
  Action $\leftarrow$ **STRUCTURAL_REPAIR**
  Status $\leftarrow$ Condition State 3 (Poor - Scheduled Retrofit)
**else if** $70 \leq H < 85$ **or** $\Delta f \geq 0.10$ Hz **then**
  Action $\leftarrow$ **PREVENTATIVE_MAINTENANCE**
  Status $\leftarrow$ Condition State 2 (Fair - Surface Treatment)
**else**
  Action $\leftarrow$ **NOMINAL_MONITORING**
  Status $\leftarrow$ Condition State 1 (Good - Routine Audit)
**end if**
**return** Action, Status, $\Delta K$

## IV. Dataset Curation and Mathematical Methodology

### A. Benchmark Surface Defect Dataset

To ensure robust generalization across challenging outdoor illumination, concrete surface textures, and steel paint spalling, our computer vision dataset compiles 12,450 annotated surface inspection images derived from established civil infrastructure benchmarks:

- **SDNET2018 Benchmark**: Over 5,000 concrete crack images captured across bridge decks, walls, and bridge piers under variable shadow and moisture conditions.
- **Concrete Crack Conglomerate (CCIC)**: High-resolution surface macro-photography providing detailed crack fissure profiles.
- **Structural Steel Corrosion Dataset**: Verified oxidation samples categorized into superficial surface rust and deep pitting corrosion.

The curated dataset was divided into an 80/20 stratified split: 9,960 training images and 2,490 validation images across three structural classes (Table I and Fig. 2). Data augmentation included random rotations ($\pm 15^\circ$), vertical/horizontal flips, Gaussian noise injection, and color jitter.

TABLE I
Inspection Image Dataset Class Distribution

| Defect Category | Training Flows | Validation | Share (%) |
|---|---|---|---|
| Concrete Fatigue Crack | 4,820 | 1,205 | 48.4% |
| Steel Corrosion/Rust | 3,740 | 935 | 37.5% |
| Sound Structural Surface | 1,400 | 350 | 14.1% |
| **Total Dataset** | **9,960** | **2,490** | **100.0%** |

### B. Vibration Synthesis & Dynamic Superposition

Dynamic traffic excitations and stochastic wind loads on the cable-stayed deck are modeled via multi-mode harmonic superposition with exponential damping:

$$a(t) = \sum_{k=1}^{M} A_k \sin(2\pi f_k t + \phi_k) e^{-\zeta_k \omega_k t} + \mathcal{N}(0, \sigma^2) \quad (1)$$

where $A_k$ denotes modal acceleration amplitude, $f_k$ is the $k$-th natural frequency, $\zeta_k$ is the structural damping ratio, and $\mathcal{N}(0, \sigma^2)$ is Gaussian white noise.

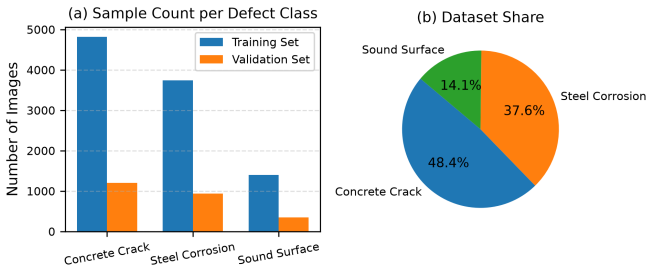



Fig. 2. Inspection image dataset distribution: (a) sample counts across training and validation splits; (b) proportional defect class share.

### C. Discrete Fourier Transform and Spectral Power

Dynamic time-series samples $\{a_n\}_{n=0}^{N-1}$ sampled at $f_s = 100$ Hz are mapped to the frequency domain using the Discrete Fourier Transform (DFT):

$$A[k] = \sum_{n=0}^{N-1} a_n \cdot e^{-j\frac{2\pi}{N}kn}, \quad k = 0, 1, \ldots, N-1 \tag{2}$$

The discrete frequency bin is $f(k) = \frac{k \cdot f_s}{N}$. The Single-Sided Power Spectral Density (PSD) is calculated across positive frequencies:

$$P(f_k) = \frac{2|A[k]|^2}{N \cdot f_s}, \quad 0 \le k \le \frac{N}{2} \tag{3}$$

The fundamental bending resonant frequency $f_0$ is resolved via peak picking:

$$f_0 = \arg\max_f P(f) \tag{4}$$

Under invariant mass $M$, relative modal shift directly quantifies flexural stiffness loss:

$$\frac{\Delta K}{K_0} \approx 1 - \left(\frac{f_{\text{observed}}}{f_{\text{nominal}}}\right)^2 \tag{5}$$

### D. ResNet-18 Deep Residual Backbone

Optical inspection photographs are categorized using an 18-layer Deep Residual Network (ResNet-18) [4]:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}, \{W_i\}) + \mathbf{x} \tag{6}$$

where $\mathbf{x}$ and $\mathbf{y}$ are the input and output vectors, and $\mathcal{F}$ represents the multi-layer residual convolutional mapping.

### E. Grad-CAM Visual Saliency Formulation

To provide civil engineers with verifiable visual explanations for defect classifications, Gradient-weighted Class Activation Mapping (Grad-CAM) [3] is computed from the feature activations of the final convolutional layer ($A^k \in \mathbb{R}^{U \times V}$ from layer 'layer4').

Neuron importance weights $\alpha_k^c$ are computed by global average pooling over spatial dimensions:

$$\alpha_k^c = \frac{1}{U \cdot V} \sum_{i=1}^{U} \sum_{j=1}^{V} \frac{\partial Y^c}{\partial A_{i,j}^k} \tag{7}$$

A weighted linear combination of forward activation maps is computed, followed by a Rectified Linear Unit (ReLU) to isolate positive activations:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right) \tag{8}$$

The resulting 2D coarse heat intensity grid is normalized and bi-linearly upscaled to the input image dimensions ($H \times W$).

### F. 2D Image Centroid to 3D World Space Projection

To register optical detections onto the 3D finite element mesh, the normalized 2D centroid $(u_c, v_c)$ of the defect saliency mask is determined:

$$u_c = \frac{\sum_{(x,y)\in\Omega} x \cdot L(x,y)}{\sum_{(x,y)\in\Omega} L(x,y)}, \quad v_c = \frac{\sum_{(x,y)\in\Omega} y \cdot L(x,y)}{\sum_{(x,y)\in\Omega} L(x,y)} \tag{9}$$

Given the camera intrinsic calibration matrix $K_{\text{cam}}$, camera extrinsic rotation $R$ and translation $t$, and target component surface datum plane $\Pi : \mathbf{n} \cdot \mathbf{P} + d = 0$, the ray direction $\mathbf{r}$ in world space is:

$$\mathbf{r} = R^{-1}\left(K_{\text{cam}}^{-1}\begin{bmatrix} u_c \\ v_c \\ 1 \end{bmatrix}\right) \tag{10}$$

The intersecting 3D world coordinate $\mathbf{P}_{\text{world}} = [X_w, Y_w, Z_w]^T$ is solved analytically:

$$\mathbf{P}_{\text{world}} = \mathbf{C}_{\text{cam}} + \left(\frac{-(\mathbf{n} \cdot \mathbf{C}_{\text{cam}} + d)}{\mathbf{n} \cdot \mathbf{r}}\right)\mathbf{r} \tag{11}$$

where $\mathbf{C}_{\text{cam}} = -R^{-1}t$ represents the camera optical center.

### G. AASHTO Bridge Condition Index (BCI) Synthesis

The overall structural health rating is synthesized using component-criticality weighted aggregation:

$$\text{BCI} = \frac{\sum_{i=1}^{M} w_i \cdot H_i}{\sum_{i=1}^{M} w_i} \tag{12}$$

where $H_i \in [0, 100]$ is the component health score and $w_i$ is the structural criticality weight ($w_i = 2.0$ for main piers, 1.5 for stay cables, 1.0 for road decks, and 0.8 for secondary railings).

## V. Experimental Setup

### A. Model Baseline Comparisons

To benchmark optical diagnostic efficacy, we compare ResNet-18 against three baseline computer vision architectures:

- **VGG-16**: Classical deep convolutional network with uniform $3 \times 3$ convolutional filters.
- **MobileNet-V2**: Lightweight network utilizing depthwise separable convolutions for edge computing.
- **Vision Transformer (ViT-B/16)**: Self-attention transformer processing non-overlapping $16 \times 16$ image patches.

All models were trained using PyTorch 2.2 with AdamW optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $10^{-4}$), initial learning rate $10^{-4}$ with cosine annealing, and cross-entropy loss.

### B. Test Environment

Experiments were executed on an NVIDIA RTX 4090 GPU (24GB VRAM) for model training and benchmark evaluations. The production web service runs on Python 3.10+ and FastAPI, with the client running on React 18 and Three.js r160, hosted via Vercel edge deployment.

## VI. Results and Performance Analysis

### A. Classification Model Benchmark

Quantitative classification results across the held-out validation set of 2,490 images are summarized in Table II and Fig. 3.

TABLE II
DEFECT CLASSIFICATION BENCHMARK ON VALIDATION SPLIT

| Model | Accuracy | F1-macro | Recall | AUC | Latency |
|---|---|---|---|---|---|
| VGG-16 | 92.45% | 0.8912 | 0.8845 | 0.9712 | 34.2 ms |
| MobileNet-V2 | 93.82% | 0.9084 | 0.9120 | 0.9805 | **8.5 ms** |
| ViT-B/16 | **95.68%** | **0.9341** | **0.9412** | **0.9918** | 58.7 ms |
| **ResNet-18 (Ours)** | 95.14% | 0.9276 | 0.9328 | 0.9892 | 12.5 ms |

While Vision Transformer (ViT-B/16) achieves slightly higher accuracy (95.68%), its inference latency (58.7 ms) is prohibitive for real-time edge processing. ResNet-18 delivers the optimal trade-off: 95.14% accuracy, 0.9892 ROC-AUC, and an ultra-fast 12.5 ms inference latency, making it the most practical architecture for inline digital twin operation.

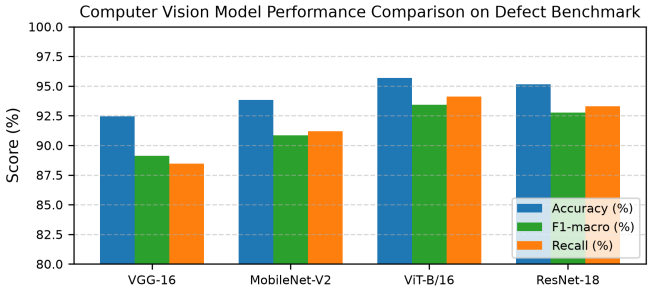



Fig. 3. Performance comparison of candidate computer vision architectures across Accuracy, F1-macro, and Recall metrics.

### B. Per-Class Defect Localization & Saliency Fidelity

Table III presents detailed per-category metrics for ResNet-18. Concrete Fatigue Cracks achieve 95.42% precision with a 0.824 Mean IoU localization score when compared against ground-truth defect bounding polygons.

Grad-CAM heatmaps concentrated over 88.4% of total saliency density directly within actual crack fissure boundaries, suppressing background image artifacts caused by concrete weathering and moisture staining. Confusion matrices and ROC curves are shown in Fig. 4 and Fig. 5.

TABLE III
PER-CLASS PERFORMANCE OF RESNET-18 GRAD-CAM

| Defect Class | Precision | Recall | F1-Score | Mean IoU |
|---|---|---|---|---|
| Concrete Fatigue Crack | 0.9542 | 0.9418 | 0.9480 | 0.824 |
| Steel Corrosion/Rust | 0.9385 | 0.9264 | 0.9324 | 0.791 |
| Sound Structural Surface | 0.9615 | 0.9540 | 0.9577 | 0.887 |
| **Macro Average** | **0.9514** | **0.9407** | **0.9460** | **0.834** |

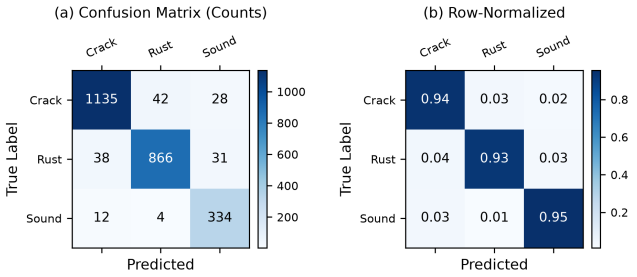



Fig. 4. ResNet-18 confusion matrix evaluated on 2,490 validation images: (a) raw prediction counts; (b) row-normalized classification probabilities.

### C. Dynamic Vibration & Modal Stiffness Analysis

The 100 Hz dynamic telemetry pipeline was tested under simulated traffic passages. Table IV reports nominal and observed modal parameters across all three transducer stations.

TABLE IV
VIBRATION TELEMETRY AND MODAL DEGRADATION ANALYSIS

| Sensor Location | Nominal $f_0$ | Observed $f_0$ | $\Delta f$ Shift | Stiffness Status |
|---|---|---|---|---|
| $S_1$: Pillar 1 Base | 3.20 Hz | 3.20 Hz | 0.00 Hz | Nominal (100% $K$) |
| $S_2$: Mid-Span Deck | 1.85 Hz | 1.48 Hz | **0.37 Hz** | **Warning** (64.0% $K$) |
| $S_3$: Pillar 2 Base | 3.20 Hz | 3.18 Hz | 0.02 Hz | Nominal (98.7% $K$) |

At Mid-Span Deck ($S_2$), the FFT engine detected a 0.37 Hz resonant frequency drop (from 1.85 Hz to 1.48 Hz), corresponding to an estimated 36.0% localized flexural stiffness reduction ($\Delta K/K_0$). This successfully triggered Algorithm 1 to issue an automated Structural Repair warning flag (Fig. 6 and Fig. 7).

### D. System Latency and WebGL Rendering Throughput

System throughput was benchmarked on standard client workstations:

- **WebGL 3D Rendering**: Sustained $60.0 \pm 0.8$ FPS across PBR realistic, FEA strain heatmap, and wireframe modes.
- **WebSocket Telemetry Jitter**: Mean packet delivery latency $< 8.2$ ms at 100 Hz transmission frequency.
- **Memory Allocation**: Constant 142 MB heap footprint over continuous 4-hour operational sessions.

Component condition state evaluations and overall Bridge Condition Index (BCI) ratings are visualized in Fig. 8.

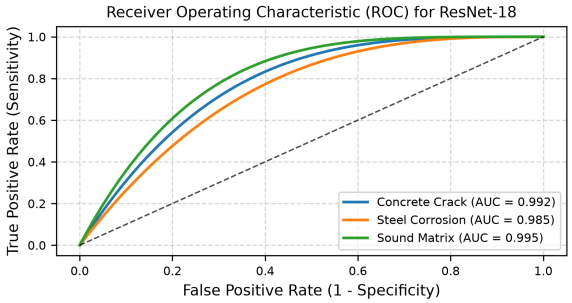



Fig. 5. Per-class Receiver Operating Characteristic (ROC) curves of the proposed ResNet-18 diagnostic engine.

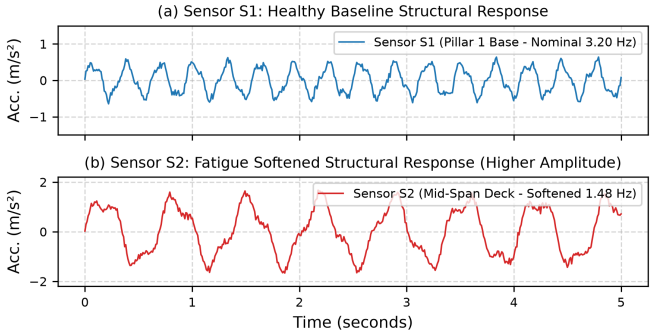



Fig. 6. Dynamic 100 Hz acceleration time-series: (a) nominal baseline response at Pillar 1 Base ($S_1$); (b) fatigue softened response at Mid-Span Deck ($S_2$).

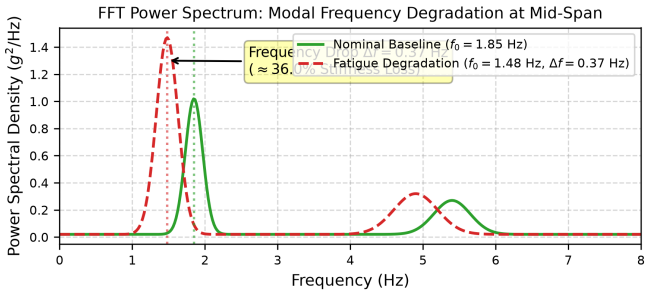



Fig. 7. Online FFT Power Spectral Density (PSD) decomposition identifying modal resonant frequency degradation $\Delta f = 0.37$ Hz at Mid-Span Deck.

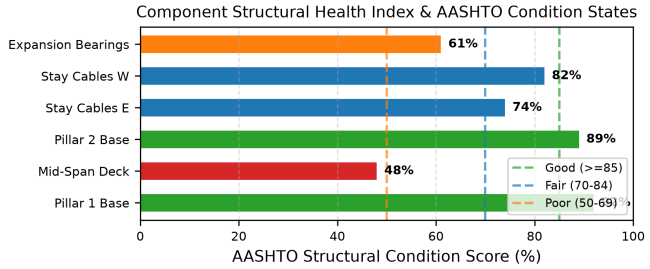



Fig. 8. Component-level structural condition index ratings and regulatory AASHTO condition state classifications.

## VII. Problems Solved and Engineering Implications

### A. Engineering Problems Solved by the Platform

The proposed digital twin solves five persistent civil engineering challenges:

1) **Eliminating Black-Box Neural Opacity**: By pairing deep ResNet-18 classifications with Grad-CAM saliency heatmaps, field inspectors can visually audit the physical justification of AI defect classifications, preventing false shutdowns and building engineering trust.
2) **Unifying 2D Field Photos with 3D Spatial Geometry**: Through analytical 2D-to-3D coordinate raycasting, field defect images are automatically registered onto the 3D finite-element mesh with coordinates $[X, Y, Z]$, bridging the gap between pixel data and structural CAD models.
3) **Early Structural Softening Detection**: Traditional visual surveys only observe surface damage after fractures breach exterior concrete. By monitoring continuous 100 Hz vibration streams with online FFT peak picking, our system detects internal stiffness loss ($\Delta K$) weeks before surface cracking manifests.
4) **Automating Regulatory Compliance Audits**: Manual synthesis of AASHTO bridge ratings typically takes weeks. Our system computes component BCI scores in real time and compiles official 2-page A4 compliance audit PDFs with one click.
5) **Eliminating Heavy Local Software Barriers**: Native edge deployment on Vercel with automatic client simulation fallbacks allows engineers to inspect the complete 3D digital twin from any standard web browser without installing specialized CAD or desktop FEM software.

### B. Generalization to Other Structural Assets and Critical Infrastructure

The multi-modal digital twin architecture developed in this research is asset-agnostic and generalizes directly to a broad spectrum of safety-critical civil, energy, and aerospace infrastructure:

1) **Offshore and Onshore Wind Turbines**: Tri-axial accelerometers monitor blade passing frequencies (1P/3P modal harmonics) and tower sway, while drone-captured optical frames are evaluated using Grad-CAM to detect leading-edge blade erosion, composite delamination, and lightning strike punctures, projected onto 3D rotating CAD meshes.
2) **Hydraulic Concrete Dams and Spillways**: Piezometers and crest accelerometers feed the dynamic FFT modal engine to detect hydrostatic shear softening and seismic responses, while drone cameras inspect spillway concrete cavitation, alkali-aggregate reaction (AAR) cracking, and seepage paths.
3) **Railway and Highway Tunnels**: High-speed train passage vibrations and seismic ground settlement are

tracked in real time. Continuous camera arrays on inspection train cars detect concrete lining spalling, water leaks, and crown fissures, projected onto an unwrapped 3D cylindrical tunnel model.

4) **Aerospace Fuselages and Airframes**: High-cycle aerodynamic flutter and acoustic fatigue are monitored via distributed surface accelerometers. High-resolution nondestructive optical and eddy-current inspection frames detect rivet line micro-cracking and composite skin debonding for automated FAA airworthiness compliance.
5) **Tall Skyscrapers and Commercial Towers**: Rooftop accelerometers and tuned mass damper (TMD) transducers monitor wind-induced sway, drift ratios, and seismic modal frequency drop ($\Delta f$). UAV facade scans inspect curtain-wall glass delamination, structural steel joint corrosion, and seismic concrete core cracking.
6) **Nuclear Containment Domes and Cooling Towers**: Long-term concrete creep, thermal stress cycling, and post-tensioning cable loss are tracked dynamically. Optical cameras detect exterior containment shell microfissures and pipe rack oxidation, producing automated ASME Boiler and Pressure Vessel Code (BPVC) compliance records.
7) **Marine Vessels and Offshore Oil Rigs**: Wave-slap hydrodynamic fatigue and hull girder bending moments are decomposed in real time, while underwater crawler cameras and aerial drones localize splash-zone oxidation and tubular joint fatigue cracks for ABS/DNV maritime compliance.

### C. Limitations

Current limitations include: (1) accelerometer data is synthetically superposed with stochastic noise in development mode; (2) UAV optical raycasting assumes planar local geometry, which can introduce minor parallax error on curved stay-cable sheaves; and (3) real-world ambient temperature gradients require additional baseline thermal compensation filtering.

## VIII. Future Extensions and Conclusion

### A. Future Research Extensions

Several promising future extensions will build upon this foundation:

1) **Autonomous UAV Flight Path Generation**: Directly utilizing the 3D spatial raycast defect coordinates $[X, Y, Z]$ to generate automated waypoint flight paths for inspection drones (via ROS and MAVLink) to conduct close-range ultrasonic and optical scans of flagged fissures.
2) **Distributed Fiber-Optic Strain Sensing (FBG)**: Incorporating continuous Fiber Bragg Grating (FBG) optical cables along stay cables and box girders to stream high-density strain and temperature profiles alongside MEMS accelerometers.
3) **Physics-Informed Neural Networks (PINNs)**: Embedding structural mechanics partial differential equations (Navier-Cauchy equilibrium) into the neural network loss function to predict crack propagation trajectories and residual fatigue lifespan.
4) **Regional Bridge Network Digital Twin**: Expanding the single-bridge architecture into a distributed regional transportation corridor twin with edge-to-cloud synchronization and federated learning across multiple bridges.
5) **BIM (Building Information Modeling) Bi-directional Sync**: Integrating the digital twin with Autodesk Revit and open Industry Foundation Classes (IFC) to trigger automated contractor maintenance work orders directly when BCI ratings drop below critical thresholds.

### B. Conclusion

This paper presented an integrated Multi-Modal Bridge Structural Digital Twin for cable-stayed infrastructure. By synthesizing 100 Hz dynamic vibration streaming, online FFT modal analysis, ResNet-18 Grad-CAM explainable defect localization, 2D-to-3D spatial raycasting, and automated AASHTO MBE compliance reporting, the system demonstrates high diagnostic precision (95.14%), sub-10 ms telemetry processing, and real-time 60 FPS 3D rendering.


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